

A hybrid Quality Diversity and Multi-objective Optimisation approach for diverse and optimised construction site layouts

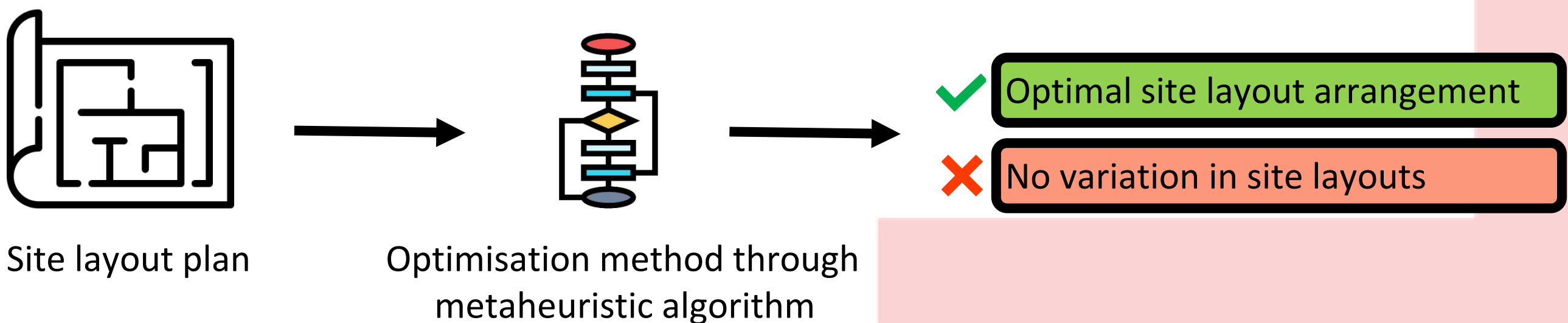
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1CHALLENGE

Construction site layout planning (CSLP) involves complex decision-making due to safety requirements, logistical constraints, and dynamic site conditions, making it difficult to balance these objectives around. Optimization methods can be used to evaluate and balance multiple objectives simultaneously. However, these methods often generate only a few “best” solutions, severely limiting the diversity of design alternatives available to planners and engineers.

Common optimization methods include metaheuristic algorithms such as genetic algorithm, particle swarm optimization, ant colony optimization etc.

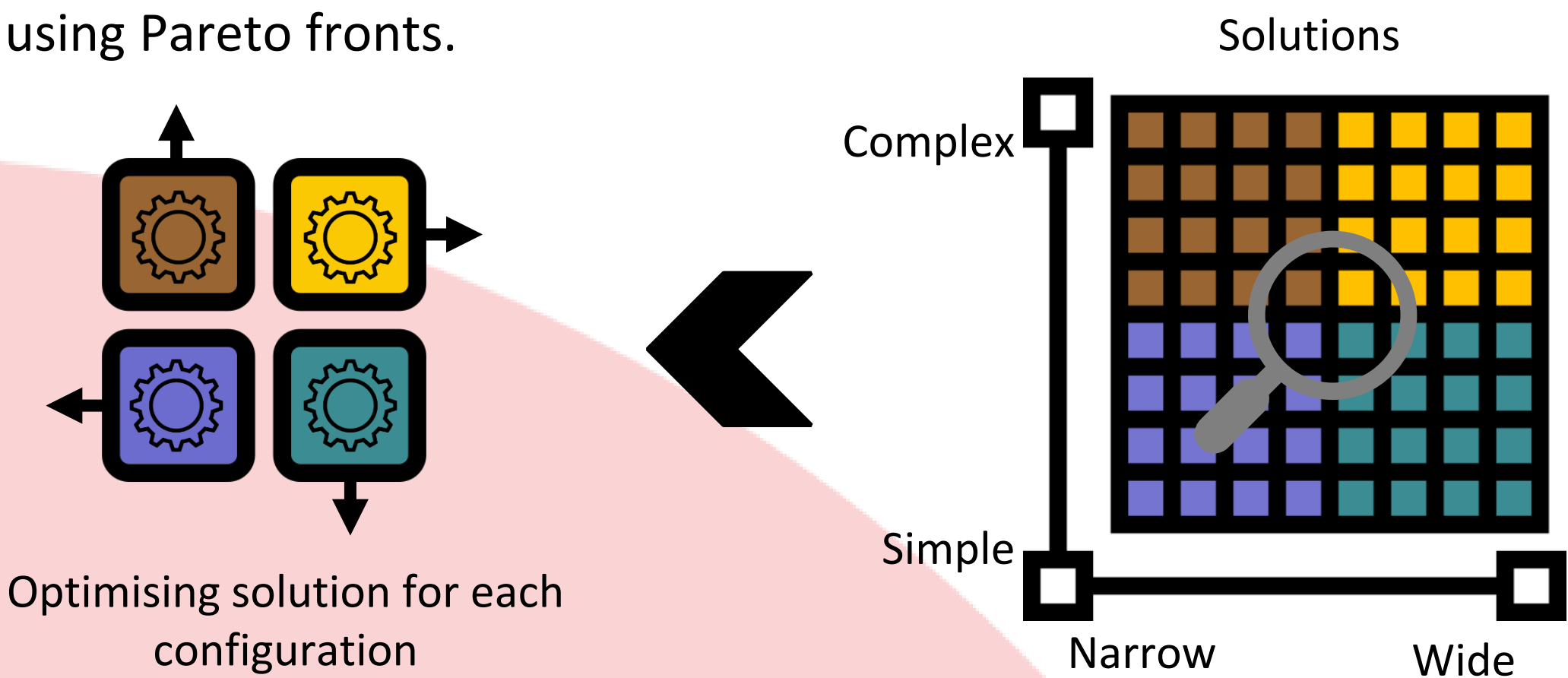


2PROPOSAL

We proposed a hybrid framework combining Quality Diversity with Multi-Objective Optimisation to generate construction layouts that are both high-performing and diverse.

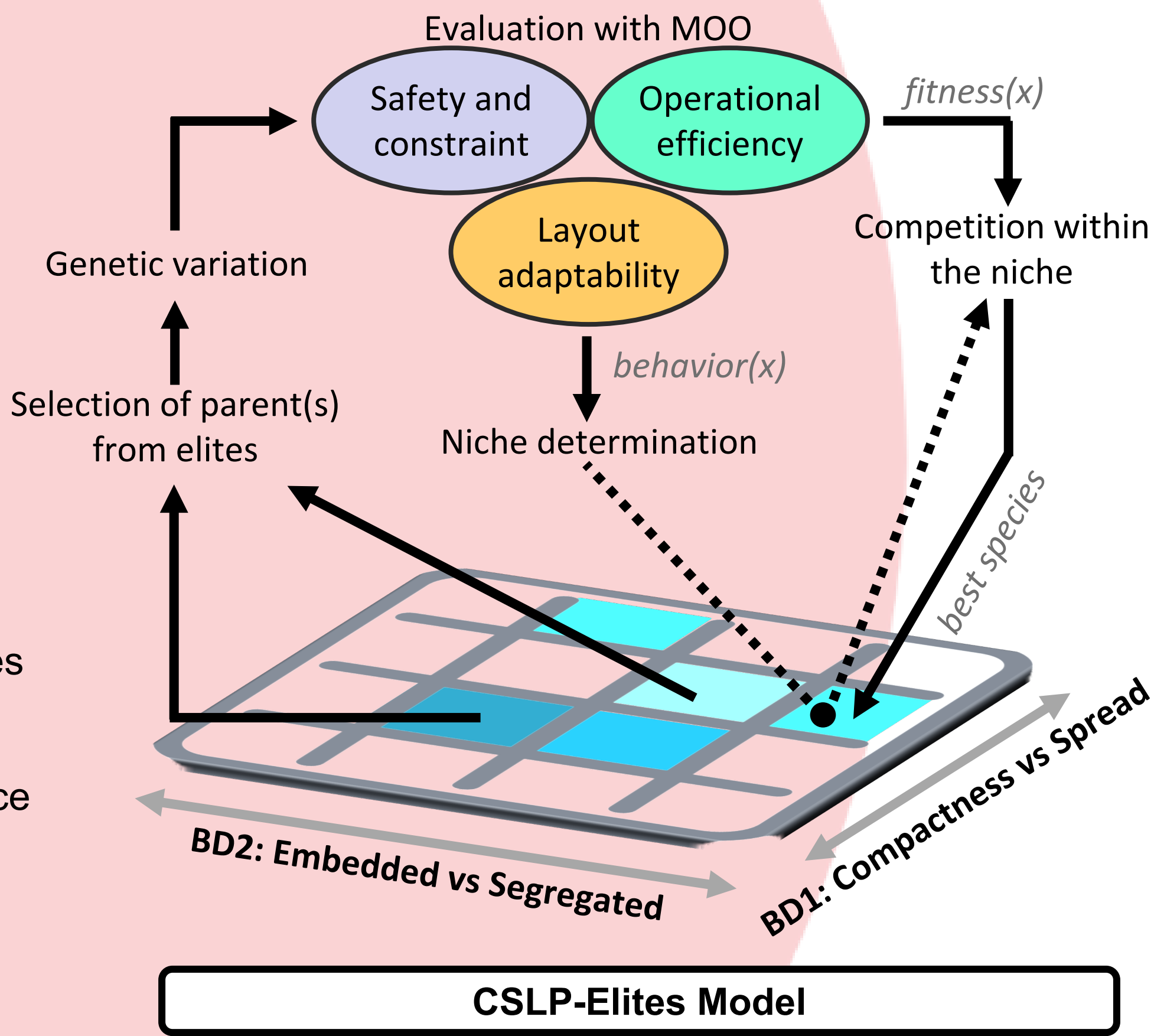
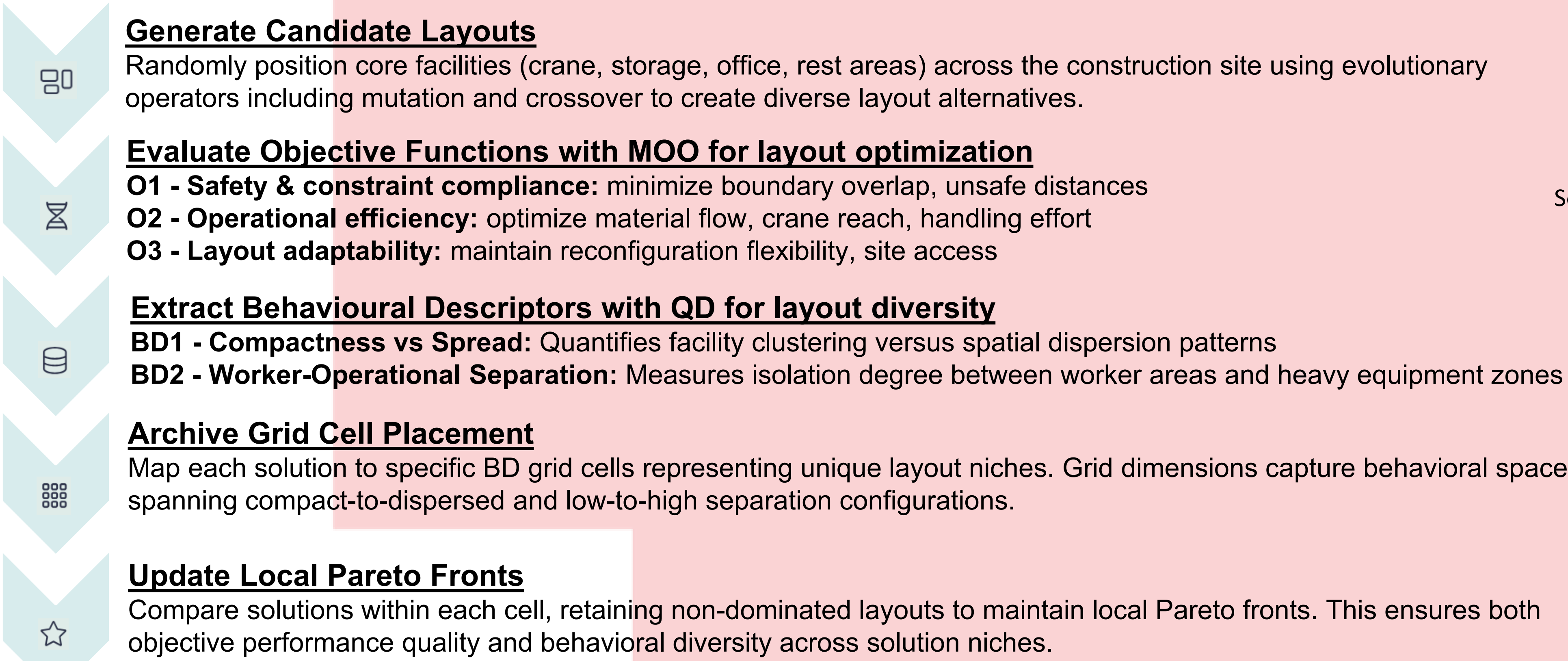
Quality diversity (QD) – Explores a wide design space to uncover many different yet feasible solutions, not just a single optimum

Multi-objective optimisation (MOO) – Identifies solutions that balance competing objectives using Pareto fronts.

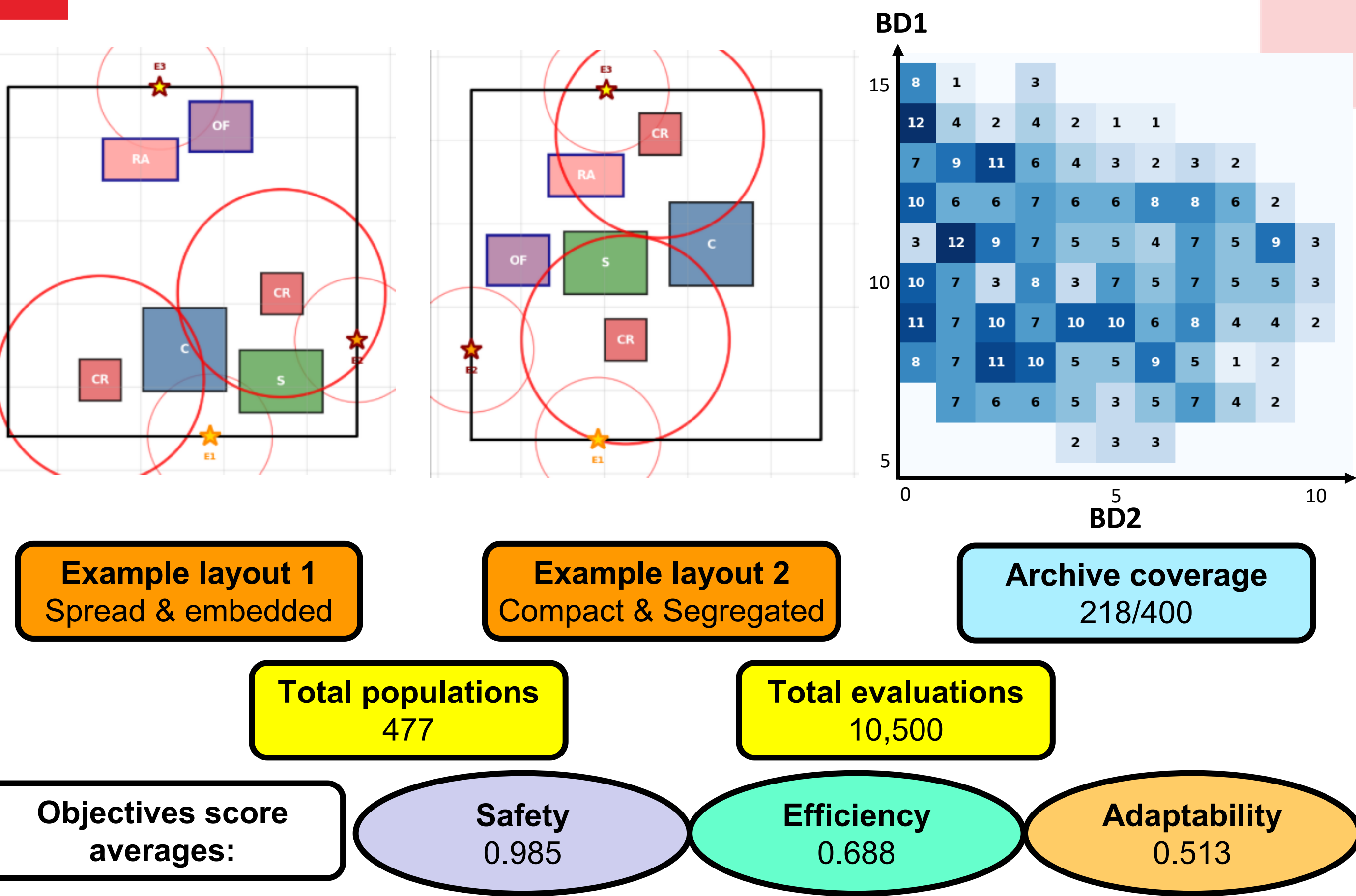


3METHODOLOGY – CSLP-ELITES MODEL FRAMEWORK

The CSLP-Elites framework employs QD optimization to generate diverse, high-performing construction site layouts through systematic behavioral characterization and archive-based selection.



4RESULTS



5CONCLUSION

This study introduced CSLP-Elites, a hybrid framework that integrates QD with MOO for CSLP. The main contributions are:

- **Hybrid framework:** Introduces a tailored integration of QD and MOO for CSLP, overcoming the limitations of conventional optimization methods that yield only a narrow set of solutions.
 - **Enhanced decision-making:** Provides planners with diverse yet high-performing layout alternatives, supporting flexibility in addressing site-specific constraints.
 - **Practical applicability:** Establishes a pathway for integration with digital construction tools such as BIM, 3D reconstruction, and simulation-based training.
- 💡 *CSLP-Elites advances construction site planning by combining optimization with diversity, enabling safer, more adaptable, and more informed design decisions.*